

Ex: Expand $f(z) = \frac{1}{z(z-1)}$ as a Laurent series for

annular domains: (1) $0 < |z| < 1$ (2) $1 < |z|$ (3) $0 < |z-1| < 1$

(4) $1 < |z-1|$.

→ We have,

$$f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

(1) For $0 < |z| < 1$

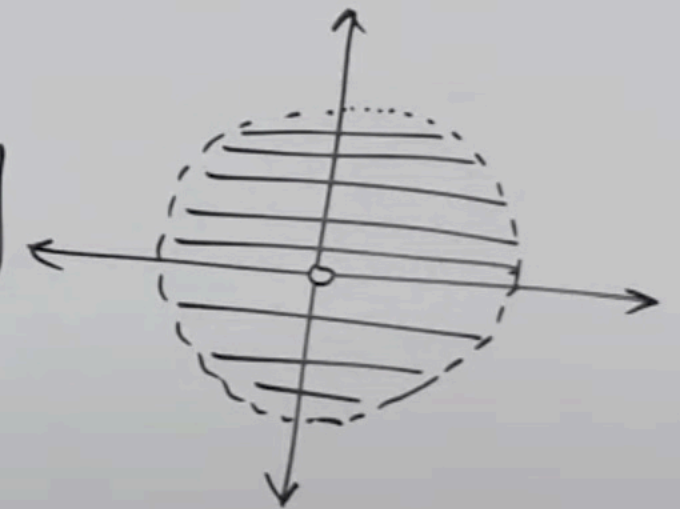
We have, $|z| < 1$

$$\begin{aligned} f(z) &= -\frac{1}{z} + \frac{1}{z-1} \\ &= -\frac{1}{z} - \frac{1}{1-z} \end{aligned}$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots, \quad |z| < 1$$

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots, \quad |z| < 1$$

$$= -\frac{1}{z} - [1 + z + z^2 + z^3 + \dots]$$



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→ We have,

$$f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

(2) For $1 < |z|$

We have, $1 < |z|$

$$\Rightarrow \left| \frac{1}{z} \right| < 1$$

$$\begin{aligned} f(z) &= \frac{-1}{z} + \frac{1}{z-1} \\ &= \frac{-1}{z} + \frac{1}{z(1-\frac{1}{z})} \end{aligned}$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots, \quad |z| < 1$$

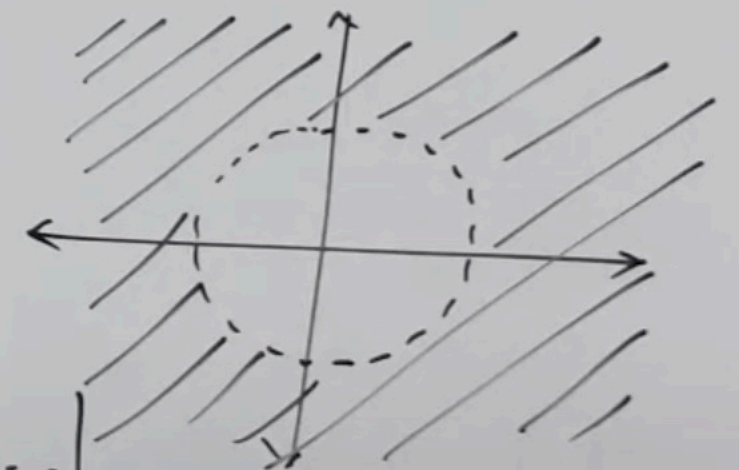
$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots, \quad |z| < 1$$

$$= \frac{-1}{z} + \frac{1}{z} \times \frac{1}{(1-\frac{1}{z})}$$

$$= \frac{-1}{z} + \frac{1}{z} \left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right]$$

$$= \frac{-1}{z} + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$$

$$= \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots$$



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(4) $1 < |z-1|$.

→ We have,

$$f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

(3) For $0 < |z-1| < 1$

We have, $|z-1| < 1$

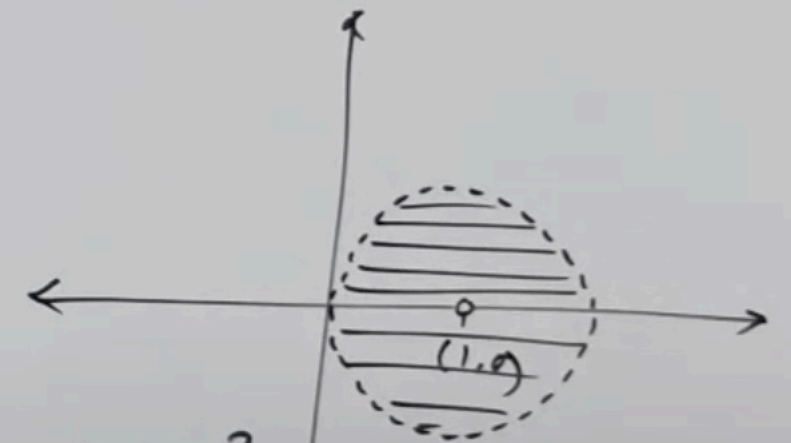
$$\begin{aligned} \text{Then } f(z) &= \frac{-1}{z} + \frac{1}{z-1} \\ &= \frac{1}{z-1} - \frac{1}{z} \\ &= \frac{1}{z-1} - \frac{1}{(z-1)+1} \end{aligned}$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots, \quad |z| < 1$$

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots, \quad |z| < 1$$

$$= \frac{1}{z-1} - \frac{1}{1+(z-1)}$$

$$= \frac{1}{z-1} - \left[1 - (z-1) + (z-1)^2 - (z-1)^3 + (z-1)^4 - \dots \right]$$



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→ We have,

$$f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + z^4 - \dots, \quad |z| < 1$$

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots, \quad |z| < 1$$

(4) For $1 < |z-1|$

We have, $1 < |z-1|$

$$\Rightarrow \left| \frac{1}{z-1} \right| < 1$$

$$\begin{aligned} \text{Then } f(z) &= \frac{-1}{z} + \frac{1}{z-1} \\ &= \frac{1}{z-1} - \frac{1}{z} \end{aligned}$$

$$= \frac{1}{z-1} - \frac{1}{(z-1)+1}$$

$$= \frac{1}{z-1} - \frac{1}{(z-1) \left[1 + \frac{1}{z-1} \right]}$$

$$= \frac{1}{z-1} - \frac{1}{(z-1)} \times \frac{1}{\left(1 + \frac{1}{z-1} \right)}$$

$$= \frac{1}{z-1} - \frac{1}{z-1} \times \left[1 - \frac{1}{z-1} + \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \dots \right]$$

$$= \frac{1}{z-1} - \frac{1}{z-1} + \frac{1}{(z-1)^2}$$

$$- \frac{1}{(z-1)^3} + \dots$$

$$= \frac{1}{(z-1)^2} - \frac{1}{(z-1)^3} + \frac{1}{(z-1)^4} - \dots$$